

Converting the Decimal Number to Binary, Octal & Hexadecimal Numbers:

SIMATS Online Programming Lab
PYTHON PROGRAMMING

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PROGRAM EDITOR

Python

Run

Save

```
1 decimal=int(input("Enter a decimal number: "))
2 binary=bin(decimal)
3 octal=oct(decimal)
4 hexa=hex(decimal)
5 print("Decimal: ", decimal)
6 print("Binary: ", binary)
7 print("Octal: ", octal)
8 print("Hexadecimal: ", hexa)
```

INPUT

25

OUTPUT

```
Enter a decimal number: Decimal: 25
Binary: 0b11001
Octal: 0o31
Hexadecimal: 0x19
```

Converting Binary, Octal & Hexadecimal Numbers to Decimal Number:

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```
1 binary = input()
2 decimal1 = int(binary, 2)
3 octal = input()
4 decimal2 = int(octal, 8)
5 hexa = input()
6 decimal3 = int(hexa, 16)
7 print("Decimal1 = ", decimal1)
8 print("Decimal2 = ", decimal2)
9 print("Decimal3 = ", decimal3)
```

INPUT

111
10
10

OUTPUT

```
Decimal1 = 7
Decimal2 = 8
Decimal3 = 16
```

Complex number:

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```
1 a = 2 + 3j
2 sum = a + 3
3 diff = a - 3
4 mul = a * 3
5 exp = a**3
6 div = a/3
7 print("Sum = ", sum)
8 print("Difference = ", diff)
9 print("Multiplication = ", mul)
10 print("Exponential = ", exp)
11 print("Division = ", div)
```

INPUT

Your INPUT goes here!

OUTPUT

```
Sum = (5+3j)
Difference = (-1+3j)
Multiplication = (6+9j)
Exponential = (-46+9j)
Division = (0.6666666666666666+1j)
```

Division:

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```
1 #Division
2 a = 1/2
3 b = 1.5/2.5
4 c = (2+3j)/(2+3j)
5 print(a)
6 print(b)
7 print(c)
```

INPUT

Your INPUT goes here!

OUTPUT

```
0.5
0.6
(1+0j)
```

Floor Division:

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Python

Run

Save

```
1 #FLOOR DIVISION
2 a = 10//2
3 b = 11.5//2.5
4 print(a)
5 print(b)
```

INPUT

Your INPUT goes here!

OUTPUT

```
5
4.0
```

Modulus:

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Python

Run

Save

```
1 #MODULUS
2 a = 15 % 2
3 b = 11.5 % 2.5
4 print(a)
5 print(b)
```

INPUT

Your INPUT goes here!

OUTPUT

```
1
1.5
```

Converting the given days into months&days:

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PROGRAM EDITOR

Python

Run

Save

```
1 #CONVERTING DAYS INTO MONTHS & DAYS
2 days = int(input())
3 months = days//30
4 rem_days = days %30
5 print(f"{months} months and {rem_days} days")
```

INPUT

125

OUTPUT

```
4 months and 5 days
```

Finding the average of 3 numbers:

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PYTHON PROGRAMMING

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192524316RD

PROGRAM EDITOR

Python

Run

Save

```
1 #AVERAGE OF 3 NUMBERS
2 a = int(input())
3 b = int(input())
4 c = int(input())
5 avg = (a+b+c)/3
6 print("Average of the given 3 numbers is", avg)
```

INPUT

```
2
3
4
```

OUTPUT

```
Average of the given 3 numbers is 3.0
```

Interchanging 2 variables using a third variable:

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PROGRAM EDITOR

Python

Run

Save

```
1 #SWAPPING TWO NUMBERS USING THIRD VARIABLE
2 a = 10
3 b = 20
4 print("Before Swapping: a = ", a, " & b = ", b)
5 temp = a
6 a = b
7 b = temp
8 print("After Swapping: a = ", a, " & b = ", b)
```

INPUT

Your INPUT goes here!

OUTPUT

```
Before Swapping: a = 10 & b = 20
After Swapping: a = 20 & b = 10
```

Interchanging 2 numbers without a third variable:

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PYTHON PROGRAMMING

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Python  ✓ Saved Run Save

```
1 #SWAPPING TWO NUMBERS WITHOUT A THIRD VARIABLE
2 a = 10
3 b = 20
4 print(f"Before: a = {a}, b = {b}")
5 a, b = b, a
6 print(f"After: a = {a}, b= {b}")
```

INPUT

Your INPUT goes here!

OUTPUT

```
Before: a = 10, b = 20
After: a = 20, b= 10
```